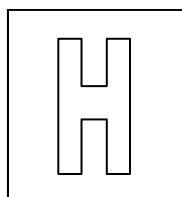


Candidate Name: \_\_\_\_\_

Class    Adm No

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## 2018 End-of-Year Exams Pre-University 2

### H1 CHEMISTRY

**8873 / 02**

Paper 2 Structured Questions

**11 Sep 2018**

**2 hours**

Candidates answer on the Question paper.

Additional materials: Data Booklet

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#### READ THESE INSTRUCTIONS FIRST

**Do not turn over this question paper until you are told to do so**

Write your name, class and admission number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| Question | 1  | 2  | 3  | 4 | 5  | Section B | Total |
|----------|----|----|----|---|----|-----------|-------|
| Marks    | 15 | 10 | 15 | 9 | 11 | 20        | 80    |

## Section A

Answer all questions in the spaces provided.

- 1 In acidic solution, bromate(V) ions,  $\text{BrO}_3^-$ , oxidises bromide ions to bromine. The progress of the reaction may be followed by adding a fixed amount of phenol (an aromatic alcohol) together with methyl red indicator.

The bromine produced during the reaction reacts very rapidly with phenol. When all the phenol is consumed, any excess bromine bleaches the indicator immediately. The initial rate of formation of  $\text{Br}_2$  is indicated by the time taken for the bromine to bleach the indicator.

- (a) An experiment was carried out four times at room temperature. The total volume of the reaction mixture is the same in all four experiments and the following data were obtained.

| Experiment | $[\text{BrO}_3^-]$<br>/ $\text{mol dm}^{-3}$ | $[\text{Br}^-]$<br>/ $\text{mol dm}^{-3}$ | Initial rate of<br>formation of $\text{Br}_2$ /<br>$\text{mol dm}^{-3} \text{ s}^{-1}$ |
|------------|----------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------|
| 1          | 0.10                                         | 0.10                                      | $8 \times 10^{-2}$                                                                     |
| 2          | 0.10                                         | 0.05                                      | $4 \times 10^{-2}$                                                                     |
| 3          | 0.05                                         | 0.05                                      | $2 \times 10^{-2}$                                                                     |
| 4          | 0.05                                         | 0.10                                      | $4 \times 10^{-2}$                                                                     |

- (i) Determine the order of the reaction with respect to each of the following reactants, showing your working clearly.

I.  $\text{BrO}_3^-$

II.  $\text{Br}^-$

[3]

Comparing Expt 1 & 2,

when  $[\text{Br}^-]$  increases by 2 times, rate increases by 2 times  $\left(\frac{8 \times 10^{-2}}{4 \times 10^{-2}}\right)$ . [1]

$\Rightarrow$  rate is directly proportional to  $[\text{Br}^-]$

$\Rightarrow$  reaction is 1<sup>st</sup> order with respect to  $\text{Br}^-$  [1]

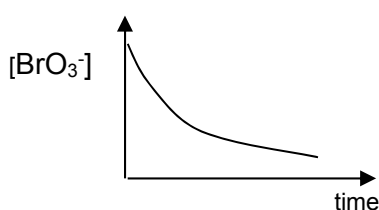
Comparing Expt 2 & 3,

when  $[\text{BrO}_3^-]$  increases by 2 times, rate increases by 2 times  $\left(\frac{4 \times 10^{-2}}{2 \times 10^{-2}}\right)$ .

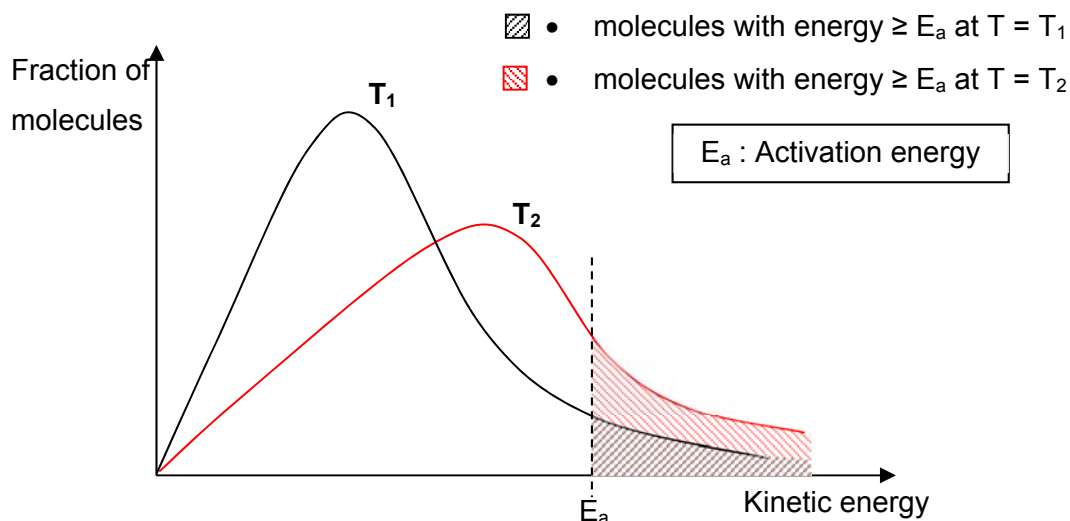
$\Rightarrow$  rate is directly proportional to  $[\text{BrO}_3^-]$

$\Rightarrow$  reaction is 1<sup>st</sup> order with respect to  $\text{BrO}_3^-$  [1]

- (ii) Based on your answer in (a)(i), sketch a concentration-time graph for  $\text{BrO}_3^-$ . [1]



- (iii) Describe, and explain in molecular terms, how the rate of reaction is affected by an increase in temperature. You should make reference to the Boltzmann distribution in your answer. [3]



When temperature increases, the average kinetic energies of the reactant molecules increase.

Thus, the number of reactant molecules with energy greater or equal to the activation energy will increase. This results in an increase in the frequency of effective collisions. Hence, the rate of reaction increases.

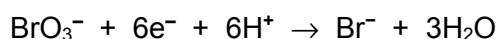
- (b) (i) State the oxidation number of the Br atom in each of the following substances:

- I.  $\text{BrO}_3^-$ ,
- II. bromide,
- III. bromine

[3]

+5, -1, 0

- (ii) In a separate experiment,  $20.0 \text{ cm}^3$  of  $0.0200 \text{ mol dm}^{-3}$   $\text{BrO}_3^-$  was found to react completely with  $80.0 \text{ cm}^3$  of  $0.0100 \text{ mol dm}^{-3}$  hydroxylamine,  $\text{NH}_2\text{OH}$ .  $\text{BrO}_3^-$  ions were reduced as shown in the ion-electron equation below.



Given that the original oxidation number of N in  $\text{NH}_2\text{OH}$  was -1, calculate the final oxidation number of N. [4]

Amount of  $\text{BrO}_3^- = 20/1000 \times 0.0200 = 0.0004 \text{ mol}$

Amount of  $\text{NH}_2\text{OH} = 80/1000 \times 0.01 = 0.0008 \text{ mol}$  ;

Amount of e gained by  $\text{BrO}_3^- = 0.0004 \times 6 = 0.0024 \text{ mol}$  ;

Mole ratio of  $\text{NH}_2\text{OH}$ : electrons =  $0.0008:0.0024 = 1:3$  ;

Final oxidation number of N =  $-1 - (-3) = +2$  ;

- (iii) Based on your answer in (b)(ii), suggest a possible product for the oxidation of hydroxylamine. [1]

NO

(ecf based on (b)(ii))

[Total: 15]

2 Magnesium and aluminium are elements from Period 3 of the Periodic Table.

- (a) State and explain **two** factors that leads to aluminium having a higher melting point than magnesium. [2]

From Mg to Al, the number of delocalised electrons and charge density of metal cation increases, thus the strength of metallic bond also increases from. Hence there is an increase in melting point from Mg to Al

- (b)  $\text{MgO}$  and  $\text{Al}_2\text{O}_3$  have giant ionic lattice structures.

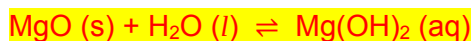
Explain why the melting point of  $\text{Al}_2\text{O}_3$  is lower than  $\text{MgO}$ . [2]

The melting point of  $\text{Al}_2\text{O}_3$  is lower than  $\text{MgO}$  due to  $\text{Al}_2\text{O}_3$  being an ionic compound with partial covalent character. This occurs as the  $\text{Al}^{3+}$  cation has a high charge density and can strongly polarise/distort the electron cloud of the  $\text{O}^{2-}$  anion.

- (c) Explain the following observations when separate samples of  $\text{MgO}$  and  $\text{Al}_2\text{O}_3$  were added to water. Write suitable equation(s) were appropriate.

I.  $\text{MgO}$  dissolves slightly in water to give a weakly alkaline solution

II.  $\text{Al}_2\text{O}_3$  did not dissolve when added to water. [2]

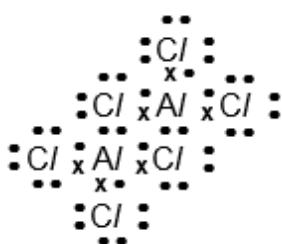


$\text{Al}_2\text{O}_3$  has high lattice energy.

- (d) Unlike  $\text{MgCl}_2$ ,  $\text{AlCl}_3$  has the tendency to dimerise to form  $\text{Al}_2\text{Cl}_6$ .

Draw the 'dot-and-cross' diagram of  $\text{Al}_2\text{Cl}_6$  and explain how the dimer was formed.

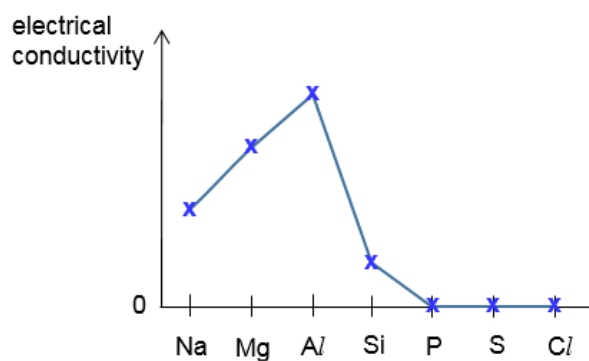
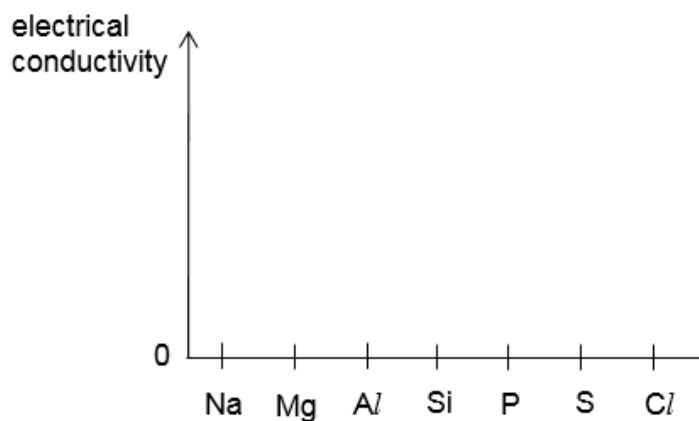
[3]



Chlorine atom possess a lone pair of electrons in its valence quantum shell.

Aluminium has energetically-accessible vacant orbitals in its valence quantum shell to accommodate the lone pair of electrons from chlorine to form a dative covalent bond.

- (e) Using the axes below, sketch the graph to show the electrical conductivity of the Period 3 elements from Na to Cl. [1]



[Total: 10]

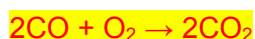
- 3 The following abstract is taken from an article from the popular online site 'Chemistry World'.

Mazda Motor Corporation has unveiled a new generation of catalytic converters that use 70% to 90% less of the precious metals which help to purify exhaust emissions. The converters rely on nanoparticles of the catalytic metal, each less than five nanometres across, studded onto the surface of tiny ceramic spheres. The Japanese firm claims this is the first time a catalyst material has been achieved that features single, nanosized precious metal particles embedded in fixed positions.

Automotive catalysts use platinum, rhodium and palladium nanoparticles instead of other larger particles to speed up chemical reactions of pollutants such as nitrogen oxide, carbon monoxide and hydrocarbons, to create non-toxic emissions. Unfortunately, using platinum and palladium nanoparticles can result in negative impacts on human health. Ongoing research suggests that emissions of platinum-group metals from catalytic converters along US highways might be a root cause of an alarming rise in allergies and asthma.

*Adapted from "Catalytic Converters Go Nano" ~ Ned Stafford*

- (a) (i) By writing a chemical equation, illustrate how automotive catalyst convert pollutants into non-toxic emissions. [1]



- (ii) Suggest why automotive catalysts use nanoparticles instead of larger particles. [2]

The higher surface area to volume ratio of platinum nanoparticles increases catalytic efficiency in catalytic converters. There are more collisions between the reactants and catalyst, increasing the frequency of effective collisions, thus speeding up the chemical reactions.

- (iii) Explain why having platinum and palladium in catalysts can translate into health risks.[1]

The small size of nanoparticles means that nanoparticles can be easily inhaled and absorbed by the lungs and subsequently absorbed into the blood stream. This can lead to health risks such as asthma, poisoning of the blood.

- (b) Gas chromatography (GC) is a type of chromatography used in analytical Chemistry for separating and analysing compounds that can be vaporised without decomposing. The composition of the pollutants in petrol in vehicles can be determined using GC. The exhaust gases are passed through a separating column and the gaseous compounds are being analysed by their interactions with the walls of the column which is coated with a stationary phase (usually a large polymer). These interactions cause the gaseous compounds to be extracted at different timings, known as the retention time of the compounds. If the polarity of the gases and the stationary phase are similar, then there is likely to be a greater interaction between the two. In other words, the retention time will be longer for polar compounds on polar stationary phases and shorter on non-polar stationary phases. Boiling points is another property that affects retention times. If a component has a low boiling point, it is likely to spend more time in the gas phase in the column instead of interacting with the walls of the column. The comparison of retention times gives GC its analytical usefulness.

The following table gives the retention timings of some pollutants.

| Compound                   | Retention time / min |
|----------------------------|----------------------|
| Hydrogen (H <sub>2</sub> ) | 3.0                  |
| Carbon monoxide (CO)       | 14.5                 |
| Nitrogen Oxide (NO)        | 18.8                 |

- (i) Account for the difference in retention times for CO and NO. [2]

Both CO and NO are simple covalent molecules. Both CO and NO has pd-pd forces of attraction between its molecules. However, NO has a larger size of electron cloud, thus more thermal energy is required to overcome the stronger pd-pd forces of attraction in NO, thus NO has a higher boiling point and a longer retention time.

- (ii) Draw a dot-and-cross diagram to illustrate the bonding in CO<sub>2</sub>.

State its shape and bond angle.

[2]

O=C=O, linear, 180°

- (c) Under room conditions and in the presence of a platinum catalyst, one mole of compound A, with molecular formula, C<sub>5</sub>H<sub>10</sub>, requires an equal amount of hydrogen gas for reduction.

- (i) Identify the functional group present in compound **A**. [1]

Alkene

- (ii) Identify the number of sigma and pi bonds in compound **A**. [2]

14 sigma, 1 pi

- (iii) Draw **and** name any **three** possible structures of **A**. [3]

Pent-2-ene / pentene / 3-methyl butene

- (iv) Compound **B** can be synthesised from **A**. Given that **B** is a ketone with a relative molecular mass of 58.0 and contains 3 carbon atoms, draw the structure of **B**. [1]

CH<sub>3</sub>COCH<sub>3</sub>

[Total: 15]

- 4 But-1-ene, CH<sub>3</sub>CH<sub>2</sub>CH=CH<sub>2</sub>, is an important raw material for the production of synthetic rubber. The enthalpy change of combustion,  $\Delta H_c$ , of but-1-ene can be determined using bond energy values.

- (a) (i) Explain the term *bond energy*. [1]

The energy required to break one mole of a covalent bond between two atoms of a molecule in gaseous state, under standard conditions of 298 K, 1 bar.

- (ii) The bond energy of the C-C bond in butane is 350 kJ mol<sup>-1</sup>. It was expected that the bond energy of the C=C bond in but-1-ene to be twice that of the C-C bond in butane. However, actual bond energy of the C=C bond in but-1-ene is only 610 kJ mol<sup>-1</sup>. Account for the difference. [2]

The C-C bond in the butane consists of 1 sigma bond. The C=C in but-1-ene consists of 1 sigma and 1 pi bond.

A pi bond is weaker than a sigma bond due to less effective overlap, hence C=C bond is less than twice of C-C bond.

- (b) (i) Write an equation to represent the standard enthalpy change of combustion of gaseous but-1-ene. [1]



- (ii) Hence, use the bond energies in the *Data Booklet* to calculate the standard enthalpy change of combustion of gaseous but-1-ene. [3]

$$\begin{aligned}\Delta H &= [2 \times \text{BE}(\text{C}-\text{C}) + \text{BE}(\text{C}=\text{C}) + 8 \times \text{BE}(\text{C}-\text{H}) + 6 \times \text{BE}(\text{O}=\text{O})] - [8 \times \text{BE}(\text{C}=\text{O}) + 8 \times \text{BE}(\text{O}-\text{H})] \\ &= (2 \times 350 + 610 + 8 \times 410 + 6 \times 496) - (8 \times 740 + 8 \times 460) \\ &= 7566 - 9600 \\ &= -2034 \approx -2030 \text{ kJ mol}^{-1}\end{aligned}$$

- (iii) In the theoretical calculations in (b)(ii), the physical state of water was assumed to be gaseous. However, in practice, **water vapour condenses into a liquid at room temperature**. State and explain how this would affect the magnitude of your answer in (b)(ii). [2]

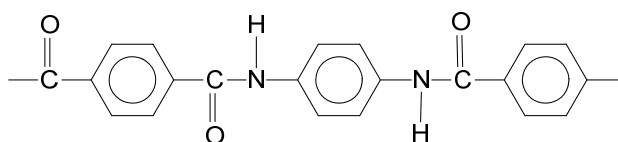
The magnitude will be larger.

When water condenses into liquid, addition energy is released.

[Total: 9]

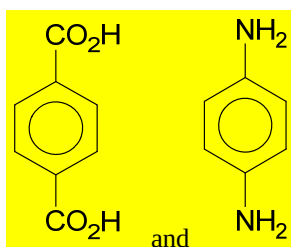
- 5 Kevlar is a heat-resistant and strong synthetic fibre used in many applications from bicycle tyres to bulletproof vests because of its high tensile strength-to-weight ratio.

- (a) (i) Part of the structure of the polymer Kevlar is shown below.



Kevlar

Draw the monomer(s) involved in the formation of Kevlar. [2]



- (ii) State the functional group present in Kevlar and state whether Kevlar is an addition or condensation polymer. [2]

Amide

Condensation polymer.

- (iii) Explain the high tensile strength of Kevlar in terms of its structure and bonding. [1]

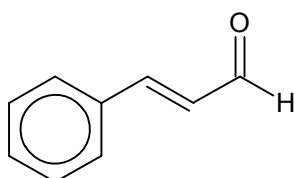
When the polymer chains are aligned, the molecules are held together by strong hydrogen bonds. / id-id

- (iv) Explain why a bullet-proof vest made of Kevlar should be stored away from concentrated acids. [1]

In the presence of acids, the amide linkages holding the monomers can be hydrolysed, thus disrupting the structure of Kevlar.

- (b) The benzene ring is an important functional group for many naturally aromatic compounds. The term was used to describe a group of compounds, many of which have aromas.

The compound that gives cinnamon its characteristic smell is cinnamaldehyde

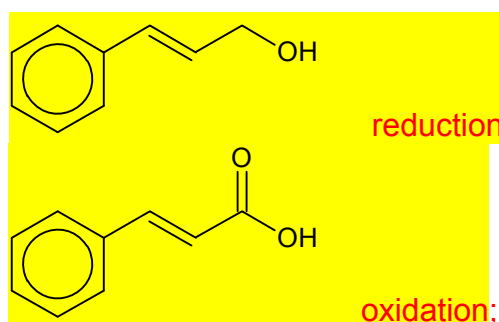


cinnamaldehyde

Draw the organic product obtained when cinnamaldehyde is treated with the following reagents:

- I.  $\text{LiAlH}_4$  in dry ether
- II.  $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ ,  $\text{H}_2\text{SO}_4(\text{aq})$ , heat

For each reagent, state the type of reaction that has occurred and record any observations observed. [5]



reduction

oxidation;  $\text{K}_2\text{Cr}_2\text{O}_7$  solution changed from orange to green.

[Total: 11]

End of Section A

## Section B

Answer any one of the two questions.

- 1 (a) Geckos are a type of reptiles that possess the ability to adhere to almost any surface. As such, research is undergoing to develop synthetic material that resembles the feet of geckos. The main reason why geckos can adhere to any surfaces is due to the nanostructures found on their feet.

For  
Examiners'  
Use

- (i) State what is the nanostructure found on geckos' feet.

[1]

**Spatulae**

- (ii) Explain how these nanostructures help geckos to stick to most surfaces.

[3]

**Spatulae forms instantaneous-dipole induced-dipole interactions with the surface. As the spatulae have high surface area to volume ratio, there is a huge cumulative surface area of contact between the gecko's feet and the surface. As such, there is a strong collective adhesive force that allows the gecko to stick on the surface.**

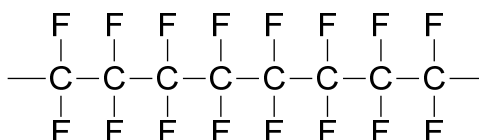
- (b) One type of materials that some geckos are unable to adhere to is polytetrafluoroethylene (PTFE). PTFE is used in making non-stick pans and other cookwares.

- (i) Define *polymers*.

[1]

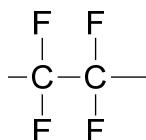
**macromolecules built up from monomers, with average molar mass of at least 1000 or at least 100 repeat units.**

- (ii) The structure of PTFE is given below.



Draw the repeat unit that make up PTFE.

[1]



- (iii) Thermoplastic and thermosetting polymers differ in their structure and bonding. Explain the differences in the rigidity and strength of thermoplastic and thermosetting polymers.

[3]

**Thermoplastics are made of linear polymers whereas thermosets are made of polymers with crosslinks.**

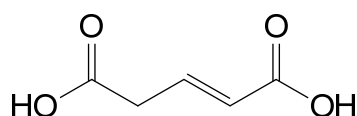
The branches of LDPE prevent maximum surface area of contact between the polymeric molecules. As a result, the instantaneous dipole-induced dipole (i.d.-i.d.) between the molecules are not extensive, less energy is required to break the i.d.-i.d. between molecules, leading to lower melting point and also greater distance of separation between the molecules. The greater distance of separation results in lower density as the mass per unit volume is smaller. The lower tensile strength arises because of the weak i.d.-i.d. too.

In general, branched-chain polymers would have a lower melting point, density and tensile strength than linear chain polymers of the same type.

- (iv) Hence or otherwise, suggest if PTFE is a thermoplastic or thermosetting polymer. [1]

**PTFE is a thermoplastic as it is a linear polymer.**

- (c) Glutaconic acid is a product of hydrolysis of proteins in the human body. Excessive levels of glutaconic acid can lead to brain damage.



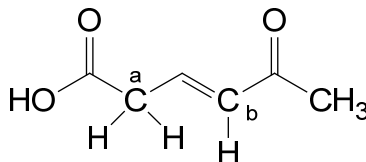
Glutaconic acid

- (i) State the IUPAC name for glutaconic acid.

[1]

**pent-2-ene-1,5-dioic acid / 2-pentene-1,5-dioic acid**

- (ii) State the bond angle and shape of the carbon stated.

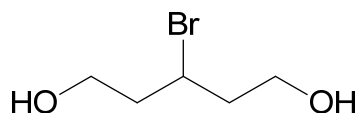


[2]

**C<sub>a</sub>: 109.5°, tetrahedral**

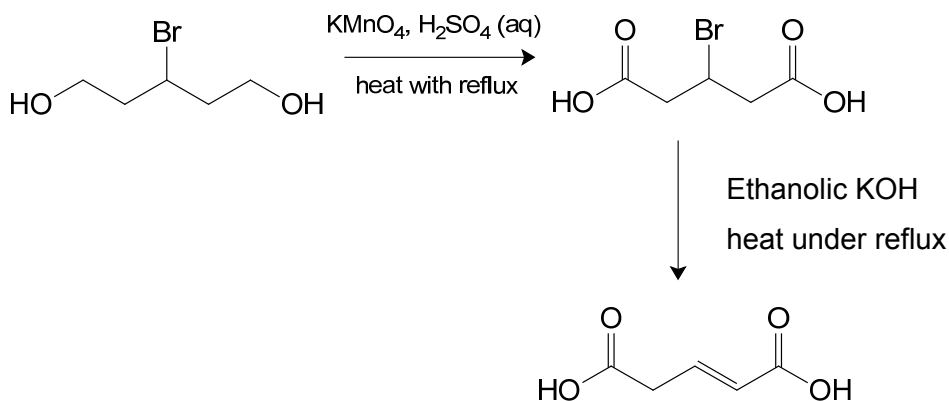
**C<sub>b</sub>: 120°, trigonal planar**

- (iii) Draw the synthetic route and suggest reagents and conditions for each step to show how glutaconic acid can be synthesised from 3-bromopenta-1,5-diol. You are required to draw the structures of all intermediates.



3-bromopenta-1,5-diol

[3]



- (iv) Describe a distinguishing test between glutaconic acid and 3-bromopenta-1,5-diol. State the observations, if any.

[2]

**Aqueous bromine. For 3-bromo-penta-1,5-diol, the orange aq. bromine will remain whereas for glutaconic acid orange aq bromine decolourises.**

- (v) Glutaconic acid is soluble in water. Explain, in terms of structure and bonding, why this happens.

[2]

**Glutaconic acid is a simple covalent molecule held together by hydrogen bonds. energy evolved from the formation of hydrogen bonds with water is sufficient enough to overcome the hydrogen bonds between the glutaconic acid molecules.**

[Total: 20]

- 2 (a) Transition metals are very useful in the chemical industries as they can act as catalysts. Vanadium, iron and nickel are commonly used as catalysts for various common reactions.

For  
Examiners'  
Use

- (i) Write out the electronic configuration of the following particles.

$\text{V}^{6+}$ : .....

$\text{Cr}^{6+}$ : .....

[2]

**$\text{V}^{6+}$ :  $1s^2 2s^2 2p^6 3s^2 3p^5$**

**$\text{Cr}^{6+}$ :  $1s^2 2s^2 2p^6 3s^2 3p^6$**

- (ii) Draw the shapes of the orbitals in which the valence electrons of  $\text{V}^{6+}$  occupy. [2]
- (iii) Define the sixth ionisation energy of vanadium. [1]
- (iv) State whether the 6<sup>th</sup> ionisation energy of vanadium is higher or lower than that of chromium. Explain your answer. [3]

**It will be higher for vanadium.**



More energy is needed to remove the electron from the 3p orbital of  $\text{V}^{5+}$  as compared to the 3d orbital of  $\text{Cr}^{5+}$  as the 3p orbital is nearer to the nucleus and hence experiences weaker screening.

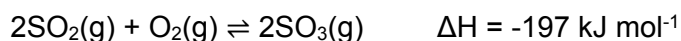
- (v) Niobium (Nb) and tantalum (Ta) are elements below that of vanadium in the periodic table. State and explain the trend in atomic radii between the three elements.

[2]

The atomic radii increase in this manner:  $\text{V} > \text{Nb} > \text{Ta}$ .

Down the group, the number of principle quantum shells increases, hence the distance between the valence electrons and nucleus increases. As such, strength of electrostatic attraction decreases and atomic radius decrease down the group.

- (b) Haber process is a very important industrial process as it is the main process involved in making of fertilisers. A similar industrial process is the Contact process, which is the process of making sulfuric acid from sulfur. The main reaction is given below.



- (i) State what is meant by Le Chatelier's Principle.

[1]

Le Chatelier's Principle states that when a system in equilibrium is disturbed, the position of the equilibrium will shift in a direction that tends to reduce that change so as to re-establish the equilibrium.

- (ii) The reaction is carried out at 450 °C and 1.5 atm. Suggest, using Le Chatelier's Principle, why this may not be the ideal condition.

[2]

The pressure is lower / close to the atmospheric pressure. If the pressure is higher, by LCP, the equilibrium position will shift towards the right so as to where there is lesser number of gaseous molecules which will help to improve the yield.

- (iii) An equilibrium was established at 700 °C in a 10 dm<sup>3</sup> vessel. The equilibrium amounts of sulfur dioxide, oxygen and sulfur trioxide were 10.0 mol, 4.0 mol and 25.1 mol.

Calculate  $K_c$  and state its units.

[3]

$$K_c = \frac{[\text{SO}_3]^2}{[\text{O}_2][\text{SO}_2]^2}$$

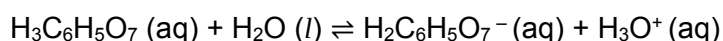
$$K_c = \frac{\left[\frac{25.1}{10}\right]^2}{\left[\frac{4.0}{10}\right]\left[\frac{10.0}{10}\right]^2} = \underline{15.8 \text{ mol}^{-1}\text{dm}^3}$$

- (iv) State the effect on the  $K_c$  calculated in (b)(iii) if the contact process is carried out at 450 °C instead of 700°C.

[1]

$K_c$  will increase

- (c) (i) Many of the food consumed commonly contain acids. One of them is citric acid, found in citrus fruits. The dissociation of citric acid is given below.



Write the  $K_a$  expression for the dissociation of citric acid.

[1]

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{H}_2\text{C}_6\text{H}_5\text{O}_7^-]}{[\text{H}_3\text{C}_6\text{H}_5\text{O}_7]}$$

- (ii) A sample of citric acid with the concentration of 0.01 mol dm<sup>-3</sup> was obtained. The pH of the solution was measured and found to be 2.54. Calculate the concentration of H<sub>3</sub>O<sup>+</sup> ions present in the solution.

[1]

$$[\text{H}_3\text{O}^+] = 10^{-2.54} = 2.88 \times 10^{-3} \text{ mol dm}^{-3}$$

- (iii) Hence, deduce if citric acid is a strong or weak acid. Explain your answer.

[1]

Citric acid is a weak acid as  $[\text{H}_3\text{O}^+]$  is smaller than that of [citric acid].

[Total: 20]

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